

COMPUTER SCIENCE TRIPOS Part IB – 2021 – Paper 4

9 Semantics of Programming Languages (nk480)

- (a) Suppose that $l : \text{intref} \in \Gamma$. Now, consider the following program equivalence for L1:

$$(\text{if } !l \leq 0 \text{ then } e_1 \text{ else } e_2); e_3 \simeq_{\text{unit}}^{\Gamma} (\text{if } !l \leq 0 \text{ then } e_1; e_3 \text{ else } e_2; e_3)$$

- (i) Explain informally but carefully why this equivalence holds. [3 marks]

Answer: The first thing both programs do is to test if the contents of l are less than or equal to 0. Depending on whether it is or not, the left-hand-side will evaluate to $e_1; e_3$ or $e_2; e_3$ – but this is exactly what the right-hand side evaluates to as well.

- (ii) Using the definition of semantic equivalence, prove that this equivalence holds. [7 marks]

Answer: First, recall that the definition of semantic equivalence says that both terms have to be well-typed, and also that for any store s such that $\text{dom}(\Gamma) \subseteq s$, both terms diverge or they evaluate to the same final configuration.

Suppose we have such an s , and now consider whether or not $s(l) \leq 0$:

- If $s(l) \leq 0$:

In this case, the configuration $\langle \text{if } !l \leq 0 \text{ then } e_1 \text{ else } e_2; e_3, s \rangle$ will step to $\langle e_1; e_3, s \rangle$ in 3 steps.

Similarly, the configuration $\langle (\text{if } !l \leq 0 \text{ then } e_1; e_3 \text{ else } e_2; e_3), s \rangle$ will step to $\langle e_1; e_3, s \rangle$ in 3 steps.

After that the two terms will have identical reduction sequences (since evaluation is deterministic), and so they will either both diverge or terminate in the same state.

- If $s(l) > 0$:

In this case, the configuration $\langle \text{if } !l \leq 0 \text{ then } e_1 \text{ else } e_2; e_3, s \rangle$ will step to $\langle e_2; e_3, s \rangle$ in 3 steps.

Similarly, the configuration $\langle (\text{if } !l \leq 0 \text{ then } e_1; e_3 \text{ else } e_2; e_3), s \rangle$ will step to $\langle e_2; e_3, s \rangle$ in 3 steps.

After that the two terms will have identical reduction sequences (since evaluation is deterministic), and so they will either both diverge or terminate in the same state.

Therefore both terms will either both diverge or terminate in the same state.

- (b) Now, consider the following *non*-equivalence:

$$e_3; (\text{if } !l \leq 0 \text{ then } e_1 \text{ else } e_2) \not\simeq_{\text{unit}}^{\Gamma} (\text{if } !l \leq 0 \text{ then } e_3; e_1 \text{ else } e_3; e_2)$$

- (i) Give a well-typed example exhibiting a counterexample of this equivalence. [5 marks]

Answer: Consider a store s maps l to 0, and $e_3 \triangleq l := 1$. Now the left-hand side will evaluate to e_2 , and the right-hand side will evaluate to e_1 , each in 5 steps.

- (ii) Give a statically decidable condition under which the transformation is valid. [5 marks]

Answer: There are many possible answers to this question. Two reasonable conditions are for e_3 to only do reads, and another one is to be typeable in the subset of Γ excluding l . As long as the proposed condition ensures that l is not modified by e_3 , it will do.
